

SUPERSTORE SALES & PROFITABILITY ANALYSIS

Week 9 Data Analyst Internship — Business Report

Dataset	Sample Superstore
Analysis Tools	Python, Pandas, Power BI
Records	9,994
Order Period	03 Jan 2014 – 30 Dec 2017

This report summarizes the end-to-end analytics workflow, business findings, and recommendations developed from the Superstore dataset.

1. Executive Summary

The analysis evaluates sales and profitability across products, categories, regions, segments and time. The dataset contains 9,994 transaction records covering 2014–2017. Total sales were \$2,297,201, generating \$286,397 in profit and an overall profit margin of 12.5%. There were 5,009 distinct orders from 793 customers, with 37,873 units sold.

Total Sales	Total Profit	Profit Margin	Orders
\$2,297,201	\$286,397	12.5%	5,009

Business Objective

The objective is to understand business performance, identify high- and low-performing areas, evaluate profitability drivers, and translate the analysis into practical recommendations for improving sales quality and profit.

2. Business Problem

Management needs a consolidated view of sales and profitability to determine which categories, regions, sub-categories and products contribute most to business performance. The analysis also examines whether discounting is associated with weaker profitability and highlights areas that may require pricing, cost or product-level review.

3. Dataset & Data Preparation

The Sample Superstore dataset contains order, customer, product, geographic, sales, quantity, discount and profit information. The analysis was performed using Python and Pandas, followed by interactive visualization in Power BI.

Data preparation step	Result
Date conversion	Order Date and Ship Date converted to datetime
Missing-value check	No missing values detected
Duplicate-row check	No duplicate rows detected
Derived fields	Year, Month, Month Name, Year-Month, Profit Margin, Shipping Days, Profit Status
Analysis outputs	Category, Region, Sub-Category, Product, Segment and Monthly summaries

4. Analysis Methodology

The analysis followed the workflow Data → Cleaning → Analysis → Visualization → Insights → Recommendations → Report. Python/Pandas was used for data preparation and exploratory analysis. Power BI was then used to present KPIs, category and sub-category performance, yearly trends, city-level results and interactive filters.

Yearly Performance

Year	Sales	Profit
2014	\$484,247	\$49,544
2015	\$470,533	\$61,619
2016	\$609,206	\$81,795
2017	\$733,215	\$93,439

Category Performance

Category	Sales	Profit
Technology	\$836,154	\$145,455
Office Supplies	\$719,047	\$122,491
Furniture	\$742,000	\$18,451

5. Key Business Insights

- **1. Technology is the strongest category by profit.** Technology generated \$145,455 in profit on \$836,154 in sales, making it the leading category in the analysis.
- **2. Furniture has the weakest profitability.** Furniture generated \$18,451 in profit despite \$742,000 in sales, indicating a comparatively low return on sales.
- **3. The West region leads regional profitability.** The West generated \$108,418 in profit and \$725,458 in sales, making it the strongest region by profit.
- **4. Copiers are the most profitable sub-category.** Copiers generated \$55,618 in profit, ahead of the other sub-categories.
- **5. Profitability improved over the analysis period.** 2017 recorded the highest annual profit at \$93,439, compared with \$49,544 in 2014.
- **6. Discounting is negatively associated with profit.** The transaction-level correlation between Discount and Profit is -0.22, indicating that higher discounts tend to coincide with lower profitability in this dataset.
- **7. Loss-making products require attention.** 301 products have negative total profit when aggregated by Product Name, creating an opportunity for product-level review.

6. Recommendations

- **Review discount strategy:** Identify high-discount transactions and products where discounting materially reduces profit. Consider tighter discount controls or targeted promotions instead of broad discounting.
- **Investigate loss-making products:** Review the products with negative total profit for pricing, shipping, sourcing and cost issues. Decisions can include repricing, supplier review, assortment changes or promotion redesign.
- **Prioritize profitable growth:** Focus sales and marketing efforts on strong categories, sub-categories and regions such as Technology, Copiers and the West region while maintaining profitability targets.

7. Power BI Dashboard

The Power BI dashboard provides an interactive summary of sales, orders, profit margin and profit, with supporting views for category, sub-category, yearly, city and trend performance. Year and Region filters allow users to explore the results dynamically.

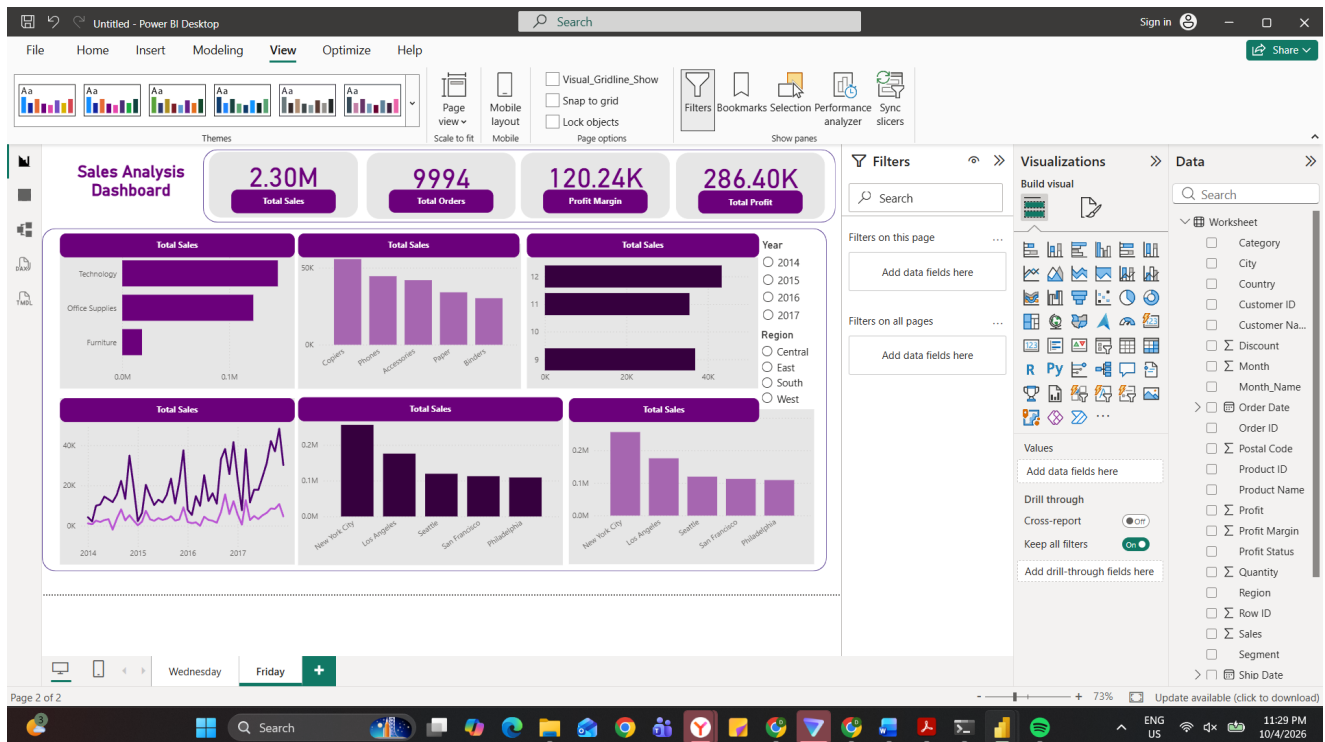


Figure 1. Wednesday Power BI dashboard draft.

8. Conclusion

The analysis provides a consolidated view of Superstore performance across 9,994 records. Overall sales of \$2,297,201 and profit of \$286,397 show a profitable business, but profitability is uneven across categories, regions and products. Technology and the West region stand out as strong performers, while Furniture and loss-making products deserve further investigation. The negative relationship between discount and profit also supports a more targeted discount strategy. These findings can be used as a foundation for the final, more polished Power BI portfolio version.

9. Project Deliverables

Deliverable	Status
Cleaned dataset	Completed
Python/Pandas analysis	Completed
Power BI dashboard	Completed — Wednesday draft
Dashboard screenshot	Included in this report
Business insights	7 insights documented
Recommendations	3 recommendations documented

GitHub repository	To be added
Professional README	To be added